Memory-Augmented Vision Transformer for Anomaly Detection

1 Problem Setup

Given a dataset of normal samples:

$$\mathcal{D}_{train} = \{x_i \sim p_{normal}(x)\} \tag{1}$$

The goal is to compute an anomaly score:

$$S(x) \uparrow \quad \text{if } x \not\sim p_{normal}(x) \tag{2}$$

2 Feature Extraction

A backbone network extracts spatial feature maps:

$$F = f_\theta(x) \in \mathbb{R}^{H \times W \times C} \tag{3}$$

Flatten into patch embeddings:

$$Q = \{q_j \in \mathbb{R}^C\}_{j=1}^N, \quad N = H \cdot W \tag{4}$$

3 Memory Bank Construction

A memory bank is constructed from normal samples:

$$\mathcal{M} = \{m_k \in \mathbb{R}^C\}_{k=1}^K \tag{5}$$

4 Soft Retrieval (Memory Conditioning)

Compute pairwise distances:

$$d_{jk} = \|q_j - m_k\|^2 \tag{6}$$

Convert distances to attention weights:

$$\alpha_{jk} = \frac{\exp(-d_{jk}/\tau)}{\sum_{l=1}^K \exp(-d_{jl}/\tau)} \tag{7}$$

where $\tau > 0$ is a temperature parameter.
Retrieve memory-conditioned features:

$$\tilde{q}_j = \sum_{k=1}^K \alpha_{jk} m_k \tag{8}$$

5 Residual (Anomaly Signal)

Compute residual vectors:

$$r_j = q_j - \tilde{q}_j \tag{9}$$

Local anomaly score:

$$s_j = \|r_j\|_2 \tag{10}$$

6 Transformer Contextualization

Construct input tokens by concatenation:

$$z_j = [q_j \| \tilde{q}_j \| r_j] \tag{11}$$

Apply a transformer encoder:

$$Z' = \text{Transformer}(\{z_j\}_{j=1}^{N}) \tag{12}$$

7 Anomaly Prediction Head

Patch-level anomaly scores:

$$\hat{s}_j = g_\phi(z'_j) \tag{13}$$

Global anomaly score:

$$S(x) = \max_j \hat{s}_j \quad \text{or} \quad S(x) = \frac{1}{N}\sum_{j=1}^{N} \hat{s}_j \tag{14}$$

8 Training Objective

Training is performed using only normal samples.

8.1 Residual Minimization

$$\mathcal{L}_1 = \frac{1}{N}\sum_{j=1}^{N} \|r_j\|^2 \tag{15}$$

8.2 Reconstruction Loss

$$\mathcal{L}_2 = \frac{1}{N}\sum_{j=1}^{N} \|q_j - g_\phi(z'_j)\|^2 \tag{16}$$

8.3 Combined Loss

$$\mathcal{L} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2 \tag{17}$$

where λ_1, λ_2 are weighting coefficients.

9 Constraints

- Memory bank is fixed during training:

$$\frac{\partial \mathcal{M}}{\partial \theta} = 0 \tag{18}$$

- Residual must remain informative to preserve anomaly signal.

10 Interpretation

The model defines an energy function:

$$E(q_j) = \|q_j - \tilde{q}_j\| \tag{19}$$

Final anomaly detection combines:

$$\text{Anomaly} = \text{Distance to Normal} + \text{Contextual Adjustment} \tag{20}$$

11 PatchCvT: Patch-Memory Conditioned Vision Transformer

PatchCvT extends PatchCore by introducing contextual transformer reasoning over memory-conditioned patch embeddings. Instead of performing purely local anomaly scoring, PatchCvT estimates semantic consistency under degradation.

11.1 Memory Retrieval

Distances to memory vectors:

$$d_{jk} = ||q_j - m_k||^2 \tag{21}$$

Soft retrieval weights:

$$\alpha_{jk} = \frac{\exp(-d_{jk}/\tau)}{\sum_{l=1}^{K} \exp(-d_{jl}/\tau)} \tag{22}$$

Memory-conditioned features:

$$\tilde{q}_j = \sum_{k=1}^{K} \alpha_{jk} m_k \tag{23}$$

Residual anomaly representation:

$$r_j = q_j - \tilde{q}_j \tag{24}$$

Energy score:

$$E_j = ||r_j||_2 \tag{25}$$

11.2 PatchCvT Tokens

$$z_j = [q_j || \tilde{q}_j || r_j || E_j] \tag{26}$$

where:

- q_j is the observed local feature
- $\tilde{q}_j$ is the retrieved normal feature
- r_j is the residual deviation
- E_j is anomaly energy

11.3 Transformer Contextualization

$$Z' = \text{Transformer}(\{z_j\}_{j=1}^{N}) \tag{27}$$

PatchCvT estimates semantic robustness confidence:

$$R_j = g_\phi(z'_j) \tag{28}$$

where R_j represents semantic consistency under degradation.

11.4 Robust Semantic Preservation

Instead of treating anomalous embeddings as automatically invalid, PatchCvT estimates whether degraded representations remain semantically useful.

Semantic preservation gating is defined as:

$$S_j = \sigma(W_r R_j - W_e E_j) \tag{29}$$

where:

- R_j is semantic robustness confidence

- E_j is anomaly energy
- W_r and W_e are learnable balancing parameters

The final robustness-conditioned representation becomes:

$$\hat{z}_j = S_j \cdot z'_j \tag{30}$$

This mechanism allows degraded but semantically coherent embeddings to remain active during downstream perception.

12 PatchCvDetr

We propose a shared-backbone architecture combining:

- a CNN backbone with memory-conditioned PatchCvT tokenization
- contextual transformer reasoning
- a DETR perception decoder

The objective is not anomaly rejection, but semantic preservation under degraded visual conditions.

PatchCvT acts as a robustness-aware regulator that prevents semantically useful degraded embeddings from being prematurely discarded by the detection transformer.

12.1 Shared Backbone

$$F = f_\theta(x) \tag{31}$$

12.2 Motion and Robustness Priors

Motion prior:

$$P_{motion} = M \tag{32}$$

Robustness prior:

$$P_{robust} = \sigma(\gamma(E - \delta)) \tag{33}$$

12.3 Attention Modulation

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + \lambda_1 P_{motion} + \lambda_2 P_{robust}\right) \tag{34}$$

where:

- P_{motion} biases attention toward dynamic regions
- P_{robust} biases attention toward degraded but semantically meaningful regions

The DETR decoder operates on robustness-conditioned embeddings:

$$\hat{Y} = DETR(\hat{Z}) \tag{35}$$

where:

$$\hat{Z} = \{\hat{z}_j\}_{j=1}^{N} \tag{36}$$

This allows the perception pipeline to preserve degraded observations that would otherwise be suppressed during attention filtering.

The detection branch predicts:

$$\hat{y} = (x, y, z, c, h) \tag{37}$$

where:

- (x, y, z) represents spatial localization
- c is class prediction
- h is an uncertainty-aware heatmap

12.4 Interpretation

The architecture combines:

$$\text{Detection} + \text{Motion Reasoning} + \text{Semantic Robustness} + \text{Contextual Normality} \tag{38}$$

Potential application domains include:

- ADAS and autonomous driving
- degraded traffic sign perception
- wheel and trajectory estimation
- sports tracking under motion ambiguity
- uncertainty-aware robotics perception